

GROUP 01

Members:

- Lê Hoàng Ân - 23125025 - Project manager
- Lê Minh Nghĩa - 23125042 - Business analyst
- Hồ Nguyễn Đăng Khoa - 23125059 - Techlead/Backend dev
- Bùi Lê Hoàng - 23125057 - UI-UX designer and front end
- Cao Hoàng Nguyễn - 23125064 - Backend dev/tester

Tên app: VietLedgr

Short description: This is a Point-of-Sale (POS) and Inventory Management System for small businesses seeking automation and intelligent insights. It features real-time stock tracking, predictive low-stock alerts, and expiry tracking for efficient stock management. An inbuilt immutable transaction ledger offers auditable, secure bookkeeping with strict role-based access. It boasts intuitive UX/UI and accessibility focused features. Using AI-driven analytics, it provides dynamic pricing, sales forecasting, and automated financial reporting — laying the groundwork for data-driven, scalable retail operations.

TL;DR: Smart retail management platform.

Target User: Small/medium business enterprise; For both owners and employees.

Timeframe: 10 weeks.

Environment: Web application (React), possible React-native implementation for mobile. Postgresql, FastAPI for backend.

Tools used:

- Moodle: used for posting and submitting assignments.

2025-2026
>
Học kỳ 1 (2025-2026)
ADVANCED PROGRAM (APCS)

Elements of Software Engineering - 23TT1

Course
Participants
Grades
Competencies

▼
General
Collapse all


Announcements

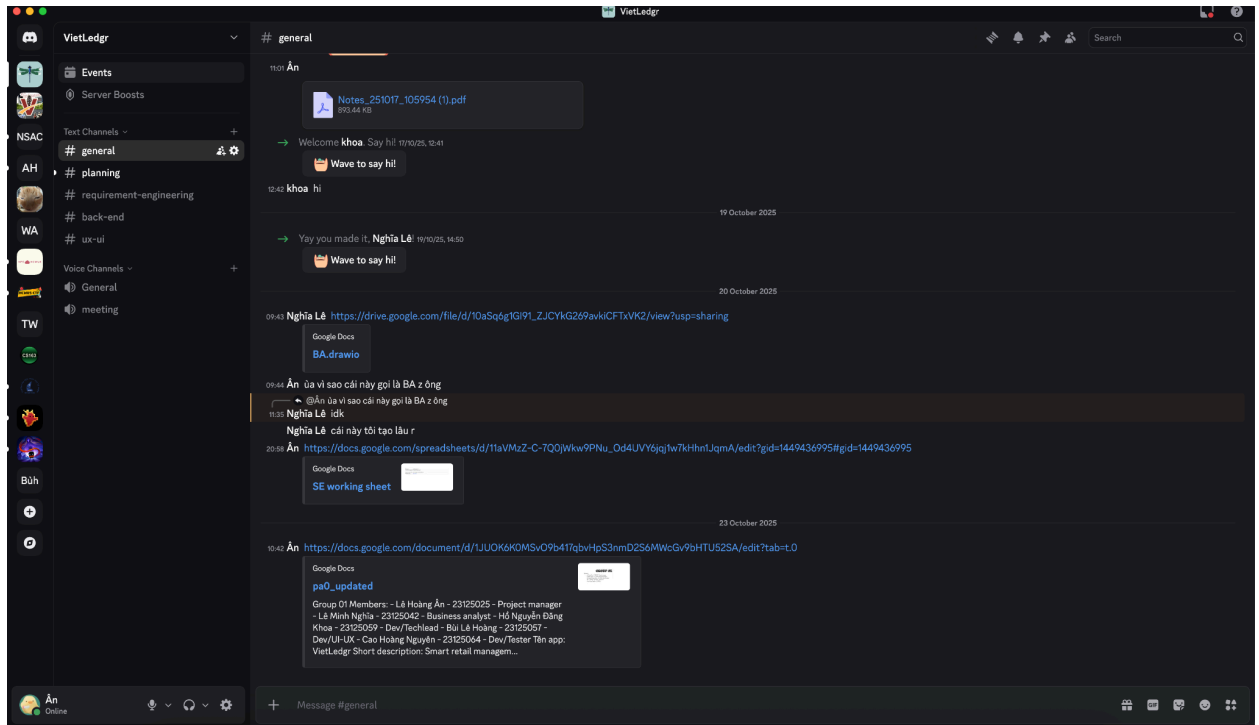
The consultation schedule of TA nndkhoa is from 11:30 AM to 1:30 PM every Wednesday, at room I82. Students have to send an email to nndkhoa@fit.hcmus.edu.vn to make an appointment in advance.

▼
Syllabus

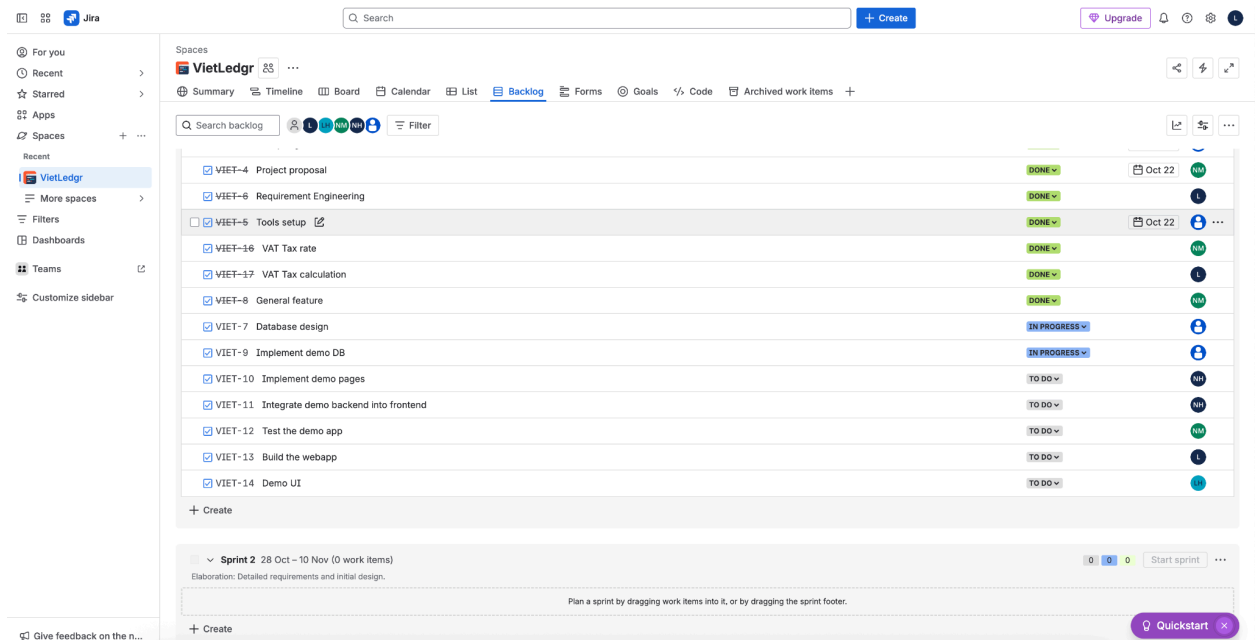

CS300-F2025-Syllabus

▼
Lecture Notes

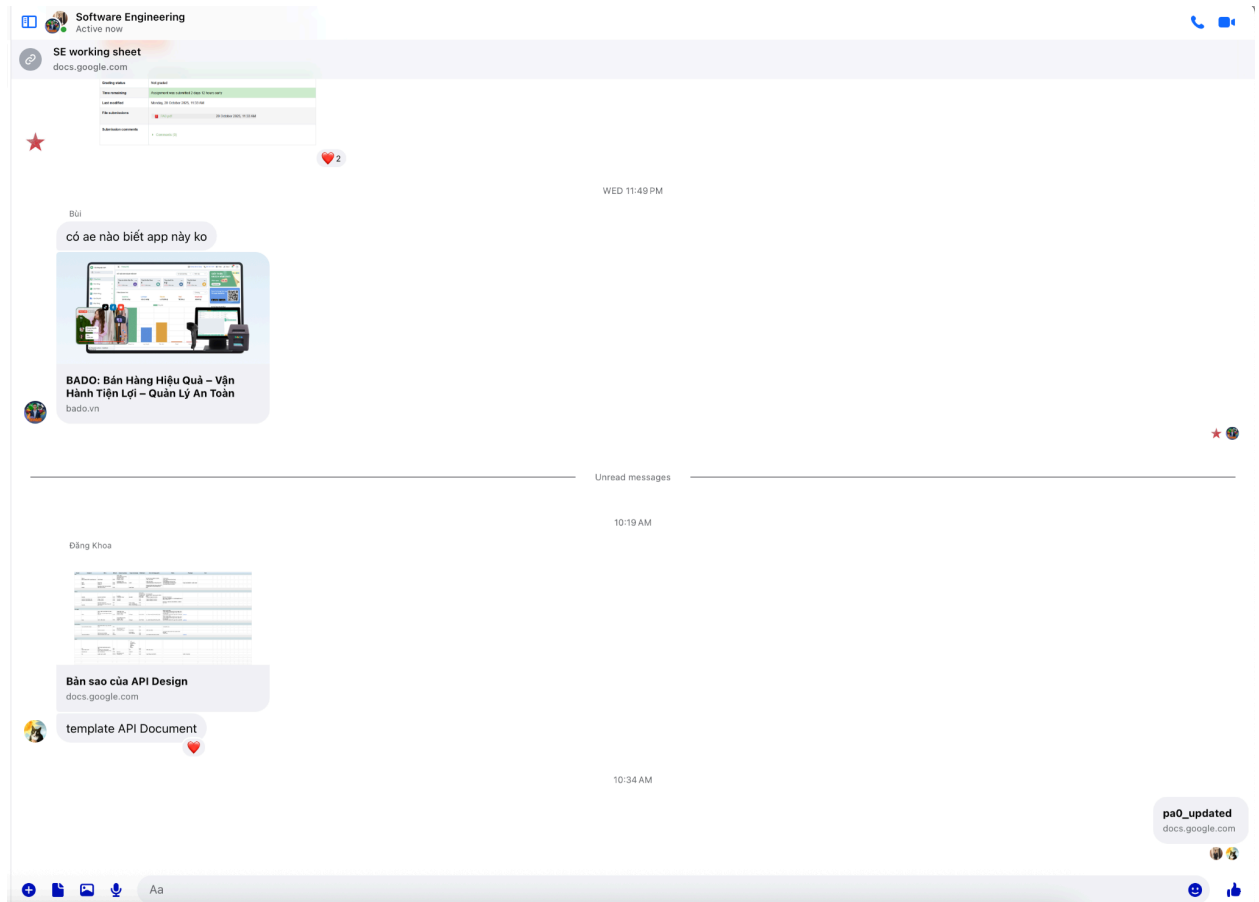
- Discord: detailed communication and meeting space.
There are channels for each activity. There is also a voice channel for meetings.
There is a fixed schedule for meetings every week (on Wednesday), the meeting must take no longer than 1 hour.



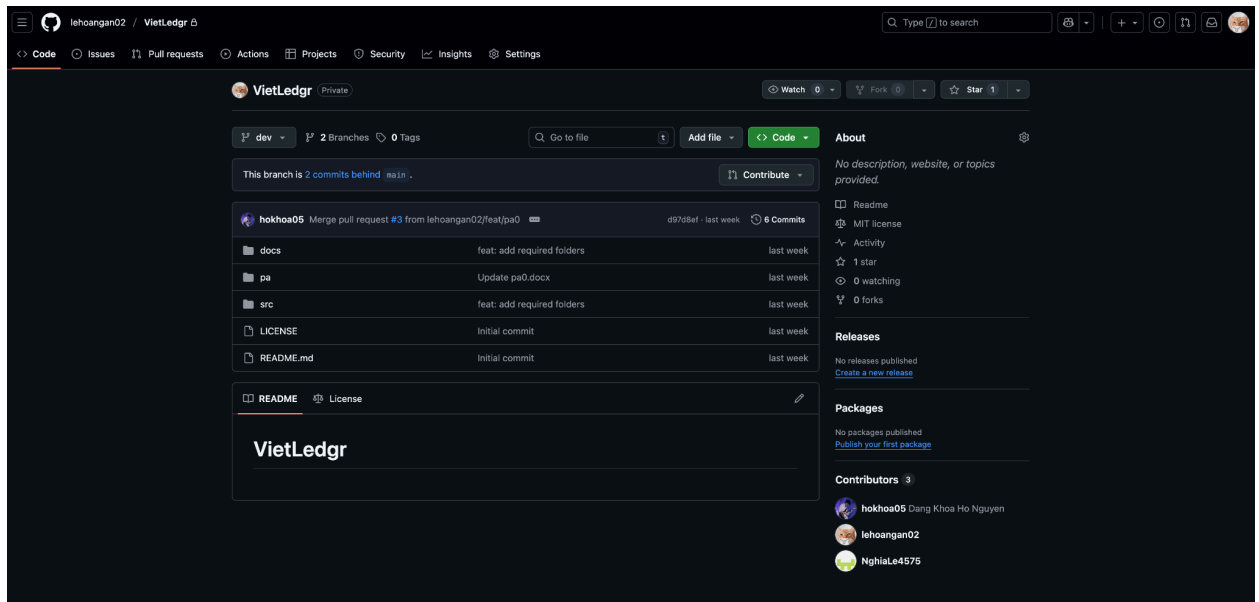
- Jira:
Jira is used to manage tasks. We group activities into sprints, following the Scrum framework.



- (Facebook) Messenger:
We use Messenger for general notifications, since every member frequently uses Messenger.



- **Github:**
We use the Git workflow to do version control. The code on the main branch must be fit for release. Each dev must have a branch for a specific feature, he/she needs to create a pull request to merge into dev. The Techlead (Khoa) needs to approve and resolve merge conflicts. After each iteration, code on the dev branch is merged into the main branch.



- Google sheet: we create a sheet to share documents.
https://docs.google.com/spreadsheets/d/11aVMzZ-C-7Q0jWkw9PNu_Od4UVY6jqj1w7kHhn1JqmA/edit?gid=0#gid=0

	A	B	C	D
1	Planning	ITERATION1.jpg		
2	Product requirement	Product Documentation Sprint 1		
3	Analysis and Design	Feature design and tax analysis	BA.drawio	
4		Updated tax analysis	Thông tin về luật thuế VAT mới nhất, cách tính thuế VAT	Thuế VAT là gì? Quy định & cách tính mới nhất Techcombank
5				

Key features:

1. Inventory & Stock Management

- **Real-time Stock Tracking:** Automatically updates when sales or restocks occur.
- **Low Stock Alerts:** Prediction of stock shortages before they happen (using sales velocity, supplier lead time).
- **Expiration Date Monitoring:** For perishable goods — system flags near-expiry items and suggests discounts or promotions to clear them.

- **(Potential): Batch & Lot Tracking:** Essential for recalls or quality control (especially if scale to enterprise clients).
- **(Potential): Supplier Integration:** Streamlined restocking workflows with supplier APIs or manual requests.

2. Transaction Ledger (Immutable Bookkeeping)

- **Immutable Ledger:** Every sale, expense, or adjustment is automatically logged with timestamp, user ID, and device ID.
- **No Manual Deletion:** Transactions can be *reversed* but never deleted or overwritten. Prevent fraud and stealing.
- **Linked Inventory Movement:** Every stock change corresponds to a financial record (no “ghost” stock loss).
- **Audit Mode:** Authorized managers can export or review every transaction chain to see *who did what, when, and why*.
- **User Roles:** Cashier, Manager, Accountant, Admin — each with explicit permissions.

3. UX/UI

- **Smart Search & Auto-Completion:** Predictive product search (by name, category, or barcode).
- **Adaptive Dashboard:** Personalized interface depending on role and behavior.
- **Minimal Click Design:** POS workflow optimized for 2–3 taps per transaction.
- **Dark/Light Modes & Accessibility Features:** WCAG-compliant for broader adoption.
- **(Potential): Voice Command Integration:** “Add 10 packs of milk” or “Show best-selling products this week.”

4. Standardisation

- **Unified Product Schema:** Standard format for product data (name, category, unit, barcode, expiration, etc.) for consistent integration.
- **Regulatory Compliance Templates:** Auto-apply VAT, GST, FDA labeling standards, etc., of the government standard.
- **(Potential) : API-First Architecture:** Easy integration with accounting software, ERP, or e-commerce platforms.
- **(Potential): International Support:** Foreign currency, tax systems, language, and units can be easily switched or added.

5. Automated Financial Reporting

- **Daily/Weekly/Monthly Summaries:** Automatically generate sales, expense, and profit reports.
- **AI-Assisted Financial Insights:** Highlight anomalies (“Sales dropped 15% this week, correlated with low stock in top items”).
- **Forecasting:** Predict next quarter’s sales or expenses using historical trends.
- **Export & Integration:** Export reports to Excel, PDF, or accounting software like QuickBooks or Xero.
- **Tax Computation:** Auto-calculate and format reports for local tax submission.

6. (Potential): QR / Barcode Integration

- **Instant Checkout:** Scan to auto-fetch product info and price.
- **QR Payment Codes:** Customer-facing QR for instant digital wallet or bank transfer payments.
- **Bulk Scanning:** Multiple item recognition for faster stock intake.
- **Smart Label Printing:** Auto-generate barcode/QR labels during inventory creation.

7. AI-driven Product & Market Insights

- **Dynamic Pricing Suggestions:** AI proposes discounts or markups to match competition or demand spikes.
- **Customer Purchase Patterns:** Suggest complementary goods (“Customers who buy X also buy Y”).
- **(Potential): Market Trend Integration:** Pull data from news feeds, social media, and e-commerce trends to highlight profitable opportunities.
- **(Potential): Product Suggestions:** Recommend new or trending products (based on news, local demand, seasonal patterns).